

Understanding root caries risk in Indonesian adults: evidence from a nationwide survey

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ABSTRACT

Background: Root caries (RC) remains a global burden and is often found in older adults. Risk factors for RC can vary in different countries. **Purpose:** This study aimed to determine the prevalence and risk factors of RC in Indonesian adults. **Methods:** This study analyzed data from Indonesia's 2018 National Health Survey, which included 37,057 respondents aged ≥ 15 years. The survey collected data on sociodemography, oral health practices, general health behaviors, and oral conditions. An oral examination was conducted to measure the number of teeth with RC, decayed, missing, or filled teeth, as well as bleeding on probing (BOP), attachment loss, pocket depth, and number of eroded teeth. Logistic regression was generated to estimate odds ratios and confidence intervals. **Results:** The prevalence of RC was 59.4%, mainly occurring in participants aged >45 years. Those aged >60 years had 2.52 times (adjusted odds ratio [AOR] = 2.52; 95% confidence interval [95%CI], 2.26–2.79) the number of RC compared with those who were younger. The likelihood of RC was increased by having a lower education (1.47 times; AOR = 1.47; 95%CI, 1.28–1.70) and never visiting a dentist (1.64 times; AOR = 1.64; 95%CI, 1.22–2.19). Decayed, missing, or filled teeth were shown to increase the risk of RC, where more coronal caries indicated a higher risk. Those with BOP (+) and attachment loss of >3 mm had a greater risk of developing RC. **Conclusion:** The prevalence of RC was relatively high, occurring mainly in middle-aged and older adults. Root caries in the Indonesian adult population reflects a complex interplay of multiple contributing factors, including sociodemography, oral health practices, general health behaviors, and oral conditions.

Keywords: Adults; epidemiology; root caries; risk factors; survey

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INTRODUCTION

Dental root caries (RC) is caries exposed on the root surface below or at the cemento-enamel junction and is diagnosed by visual and tactile methods. Root caries is caused by demineralization due to fermenting carbohydrates by cariogenic bacteria, which produce acid on the exposed root surface. Studies show that RC is commonly found in elderly people and has become more prevalent as average life expectancy has increased globally.¹ Several countries have reported a high prevalence of RC in people aged ≥ 45 years in the last 10 years, including Mexico (96%), Greece (38%), the United States (25%), Australia (77%), and

Denmark (75%).² One meta-regression analysis reported an annual incidence and increment of decayed/filled root surfaces of 32.95% and 0.64, respectively, in follow-up studies lasting <2 years, whereas these values were 9.4% and 0.43 in studies with follow-up lasting >5 years.³

As life expectancy increases and access to dental services improves globally, the number of natural teeth retained in the oral cavity is expected to increase. On the other hand, in elderly people, the risk of developing oral diseases is increasing due to systemic diseases and decreased organ function, including coronal caries, RC, and periodontal disease.^{3,4} As RC can cause pain and loss of teeth, it can significantly affect the quality of life.⁵

According to Indonesia's 2018 National Health Survey,⁶ individuals aged ≥ 3 years reported an RC prevalence of 56.6%, with the highest prevalence of 75.6% in the age range of 35–54 years. The RC index was 2.6, and the highest index was found in those aged >55 years (4.4). The average number of normal roots (i.e., exposed roots without caries) was 5.5, and the average number of root fillings (RF) was almost zero in all age categories. The ratio of root decay (RD) to normal roots was 1:2. These data show that RC remains a problem that must be addressed by the Indonesian government.

A review addressing the epidemiology of RC reported several associated risk factors, including age, gender, exposed root surfaces, coronal caries, residence status, tobacco use, sugar consumption, tooth-brushing behavior, denture wearing, streptococcus count, lactobacilli count, and xerostomia.⁷ Other studies have also found betel-chewing, periodontitis, erosion, and fluorosis to be associated with root and coronal caries.⁸ Therefore, the varying prevalence of caries reported by several countries may be attributed to their different cultures, behaviors, health programs, and, thus, risk factors.

Currently, the Indonesian population pyramid is dominated by young adults, meaning that in the next decade, the elderly population will increase.⁹ Although research related to RC in Indonesia remains limited, the data obtained from the 2018 national survey can be used to explore the factors that are associated with the occurrence of RC in the community. This study aims to describe the prevalence of, and risk factors for, RC in Indonesian adults based on sociodemographic information, oral health practices, general health behaviors, and oral conditions taken from national survey data.

MATERIALS AND METHODS

This study analyzes data from the 2018 Indonesian National Health Survey conducted by the Indonesian Health Development Policy Agency, which can be accessed with specific requirements and procedures via www.badankebijakan.kemkes.go.id.⁶ The survey was conducted at the national level, with a cross-sectional and non-interventional design. Ethical approval for the study was obtained from the Committee of Health Research Faculty of Medicine, Universitas Islam Negeri Maulana Malik Ibrahim Malang (No. 21/EC/KEPK-FKIK/40/VIII/2023).

The national survey's target sample consisted of 30,000 census blocks derived from the National Socioeconomic Survey conducted by the Central Bureau of Statistics. Sampling was performed using a probability-proportional-to-size approach combined with systematic, linear, two-stage sampling. In the first stage, all census blocks from the 2010 Population Census were implicitly stratified according to welfare strata. In the second stage, 10 households were chosen within each census block through updated systematic sampling, with further implicit stratification based on the

highest educational attainment of the household head to ensure diverse representation. Each participant provided consent by signing an informed consent form and received compensation from the government for their participation.⁶ The study included people aged ≥ 15 years, with categories of young adults (15–44 years), middle adults (45–59 years), and older adults (≥ 60 years)¹⁰; edentulous participants were excluded. Finally, the number of samples analyzed was 37,057.

The national survey employed a combination of structured interviews and oral examinations. Interviews utilized standardized questionnaires to obtain detailed socio-demographic and behavioral information. Sociodemographic variables encompassed place of residence, sex, age, and educational attainment. Behavioral variables included oral health practices—such as consumption of sweets and carbonated beverages, toothbrushing frequency, denture use, betel chewing, and dental attendance within the preceding year—as well as general health behaviors, including tobacco use, alcohol consumption, and regular medication intake.⁶

Oral examinations were performed using the World Health Organization (WHO)-stipulated form and protocol. Instruments for the oral examination included plane mouth mirrors, metallic periodontal probes, Community Periodontal Index probes conforming to WHO specifications, and several pairs of tweezers.⁶ Data on RC, decayed, missing, or filled teeth (DMF-T), bleeding on probing (BOP), attachment loss, pocket depth, and tooth erosion were obtained from the oral examinations. Root caries is identified by observing and counting the presence of RD and RF. The value for RD corresponded to the number of exposed roots with unfilled caries, whereas the value for RF corresponded to the number of exposed roots with filled caries. The DMF-T index was used to count the number of coronal caries based on category (i.e., decayed, missing, or filled). The BOP data were obtained by looking at the bleeding in the gingiva 1 minute after walking probing was performed. Gingival bleeding was recorded with scores of 0 (no bleeding) and 1 (bleeding). Loss of attachment was obtained by examining the index teeth (namely teeth 17/16, 11, 26/27, 47/46, 31, and 36/37) and was measured based on the highest score when probing using the following scoring system: 0 (0–3 mm), 1 (4–5 mm), 2 (6–8 mm), 3 (9–11 mm), and 4 (≥ 12 mm). Pocket depth was measured by walking probing and recorded as scores of 0 (no pocket), 1 (pocket = 4–5 mm), and 2 (pocket ≥ 6 mm). Tooth erosion was determined by observing the damage to the hard tissue of the teeth, which typically looks smooth and shiny, occurring in the enamel with or without involving dentin, and is not caused by a bacterial infection. Tooth erosion severity was determined by the area involved (from enamel to pulp), as recorded with the following criteria: 0 (no erosion), 1 (erosion of the enamel), 2 (erosion of the dentine), and 3 (erosion has reached the pulp).⁶ The four categories were merged into two categories: “yes” for scores of 1–3 and “no” for scores of 0.

To ensure data quality, recruitment, selection, and training procedures were implemented for both interviewers and examiners. Interviewers were required to possess at least a health science degree, and oral assessments were exclusively conducted by licensed dentists. The Indonesian Health Research Association conducted an independent external validation of the study.⁶

The data were analyzed by univariate, bivariate, and logistic regression using IBM SPSS version 28.0 (IBM,

Armonk, New York). A chi-squared analysis was performed to examine the differences between individuals with and without RC as the dependent variable based on different independent variables, including sociodemography, oral health practices, general health behaviors, and oral conditions. A logistic regression with the enter method was performed to estimate adjusted odds ratios (AORs) and 95% confidence intervals (95% CIs) for independent and dependent variables.

Table 1. Respondent characteristics and prevalence of root caries based on sociodemography, oral health practice, general health behavior, and oral condition

Variable	Cut off	n	%	Root caries				p-value
				No		Yes		
				n	%	n	%	
Age (years)	15–44 (young adult)	19,962	53.9	10,363	51.9	9,599	48.1	<0.001*
	45–60 (middle adult)	11,109	30.0	3,153	28.4	7,956	71.6	
	>60 (older adult)	5,986	16.1	1,545	25.8	4,441	74.2	
Gender	Female	21,874	59.1	9,039	41.3	12,835	58.7	<0.001*
	Male	15,183	40.9	6,022	39.7	9,161	60.3	
Location	Urban	18,772	50.7	8,200	43.7	10,572	56.3	<0.001*
	Rural	18,285	49.3	6,861	37.5	11,424	62.5	
Formal education	College	2,266	6.1	1,195	52.7	1,071	47.3	<0.001*
	High school	15,673	42.3	7,559	48.2	8,114	51.8	
	Elementary	10,958	29.6	3,817	34.8	7,141	65.2	
	No school	8,160	22	2,490	30.5	5,670	69.5	
Wear dentures	No	23,249	62.7	8,331	35.8	14,918	64.2	<0.001*
	Yes	497	1.3	184	37.0	313	63.0	
Visit dental professional in the last year	>7 times	368	1.0	181	49.2	187	50.8	<0.001*
	4–6 times	1,100	3.0	507	46.1	593	53.9	
	1–3 times	4,297	11.6	1,712	39.8	2,585	60.2	
	Never	31,292	84.4	12,661	40.5	18,631	59.5	
Eat sweet	Never	1,993	5.4	744	37.3	1,249	62.7	<0.001*
	<3 times a month	3,929	10.6	1,565	39.8	2,364	60.2	
	1–2 times a week	9,794	26.4	3,893	39.7	5,901	60.3	
	3–6 times a week	7,912	21.4	3,272	41.4	4,640	58.6	
	1 time a day	7,261	19.6	2,939	40.5	4,322	59.5	
	>1 time a day	6,168	16.6	2,648	42.9	3,520	57.1	
	Never	2,002	5.4	826	41.3	1,176	58.7	
Drink sweet	<3 times a month	2,193	5.9	920	42.0	1,273	58.0	<0.001*
	1–2 times a week	5,677	15.3	2,391	42.1	3,286	57.9	
	3–6 times a week	5,363	14.5	2,343	43.7	3,020	56.3	
	1 time a day	10,560	28.5	4,200	39.8	6,360	60.2	
	>1 time a day	11,262	30.4	4,381	38.9	6,881	61.1	
Drink soda	No	27,356	73.8	10,476	38.3	16,880	61.7	<0.001*
	Yes	9,701	26.2	4,585	47.3	5,116	52.7	
Daily toothbrushing	Yes	35,378	95.5	14,483	40.9	20,895	59.1	<0.001*
	No	1,679	4.5	578	34.4	1,101	65.6	
Take regular medication	No	10,711	28.9	4,160	38.8	6,551	61.2	<0.001*
	Yes	13,035	35.2	4,355	33.4	8,680	66.6	
	No	24,729	66.7	10,496	42.4	14,233	57.6	
Smoking	Not everyday	2,972	8.0	1,228	41.3	1,744	58.7	<0.001*
	Everyday	9,356	25.2	3,337	35.7	6,019	64.3	
Chew betel	No	36,228	97.8	14,710	40.6	21,518	59.4	<0.001*
	Yes	829	2.2	351	42.3	478	57.7	
Drink alcohol in the last month	No	36,347	98.1	14,662	40.3	21,685	59.7	<0.001*
	Yes	710	1.9	399	56.2	311	43.8	
	Free caries	36,46	9.8	3,497	95.9	149	4.1	
Decayed, missing, or filled teeth	Very low	10,942	29.5	6,030	55.1	4,912	44.9	<0.001*
	Low	8,578	23.1	2,708	31.6	5,870	68.4	
	Moderate	6,046	16.3	1,316	21.8	4,730	78.2	
	High	7,845	21.2	1,510	19.2	6,335	80.8	
Bleeding on probing (+)	No	11,235	30.3	5,642	50.2	5,593	49.8	<0.001*
	Yes	25,822	69.7	9,419	36.5	16,403	63.5	
Pocket depth > 3 mm	No	24,214	65.3	10,990	45.4	13,224	54.6	<0.001*
	Yes	12,843	34.7	4,071	31.7	8,772	68.3	
Attachment loss > 3 mm	No	24,934	67.3	15,061	40.6	21,996	59.4	<0.001*
	Yes	12,123	32.7	3,345	27.6	8,778	72.4	
Erosion	No	32,621	88.0	13,712	42.0	18,909	58.0	<0.001*
	Yes	4,436	12.0	1,349	37.0	3,087	69.6	
Total		37,057	100	15,061	40.6	21,996	59.4	

*Chi-squared, significant ($p < 0.05$)

RESULTS

Table 1 describes the prevalence of RC in different groups, which reached 59.4%. According to the chi-squared test, significant differences (with p -values of <0.05) were observed across all variables. Descriptively, it can be seen that increased age correlated with a higher prevalence of RC, with people aged >45 years showing a prevalence of $>70\%$. The prevalence of RC was higher in men (60.3%) than in women (58.7%). Root caries was more often observed in participants who lived in rural areas (62.5%), had lower formal education (69.5%), did not wear dentures (64.2%), and never visited the dentist

(59.5%). Those who never ate sweets were more likely to have RC (62.7%), whereas those who consumed more than one sweet drink per day tended to have RC (61.1%). Participants who drank soda (52.7%), did not brush their teeth (65.6%), or took medicine regularly (66.6%) tended to have higher levels of RC than their counterparts. In terms of smoking behavior, respondents who smoked every day (64.3%), never chewed betel (59.4%), or did not drink alcohol every day (59.7%) had higher levels of RC than their counterparts. Increased prevalence of RC was also seen in those with high levels of DMF-T (80.8%), BOP (+) (63.5%), pocket depth (68.3%), attachment loss (72.4%), and erosion (69.6%).

Table 2. Logistic regression analysis of the determinants that influence root caries

Variables		p -value	Odds ratio	95%CI	
				Lower	Upper
Age (years)	15–44 (young adult)	reference	-	-	-
	45–60 (middle adult)	$<0.001^*$	1.78	1.62	1.96
	>60 (older adult)	$<0.001^*$	2.52	2.26	2.79
Gender	Female	reference	-	-	-
	Male	0.68	0.98	0.89	1.08
Location	Urban	reference	-	-	-
	Rural	$<0.001^*$	1.16	1.09	1.24
Education level	College	reference	-	-	-
	High school	$<0.001^*$	1.38	1.21	1.56
	Elementary	$<0.001^*$	1.53	1.34	1.75
Wear denture	No school	$<0.001^*$	1.47	1.28	1.70
	No	reference	-	-	-
Visit dental professional in the last year	Yes	$<0.001^*$	0.59	0.48	0.73
	>7 times	reference	-	-	-
	4–6 times	0.37	1.16	0.84	1.61
Eat sweets	1–3 times	0.06	1.32	0.98	1.78
	Never	$<0.001^*$	1.64	1.22	2.19
	Never	reference	-	-	-
Consume sweet drinks	<3 times a month	0.94	1.00	0.85	1.19
	1–2 times a week	0.90	0.99	0.85	1.15
	3–6 times a week	0.83	0.98	0.84	1.15
	1 time a day	0.49	1.05	0.90	1.23
	>1 time a day	0.18	0.89	0.76	1.05
Drink soda	Never	reference	-	-	-
	<3 times a month	0.55	0.94	0.78	1.13
	1–2 times a week	0.61	0.96	0.82	1.12
	3–6 times a week	0.89	0.98	0.84	1.16
	1 time a day	0.86	1.01	0.87	1.17
Daily toothbrushing	>1 time a day	0.59	0.96	0.82	1.11
	No	reference	-	-	-
Take medication regularly	Yes	$<0.001^*$	0.88	0.81	0.94
	No	reference	-	-	-
Smoking	Yes	0.02*	0.81	0.68	0.96
	No	reference	-	-	-
Chew betel	Yes	$<0.001^*$	1.20	1.12	1.28
	No	reference	-	-	-
Drink alcohol in the last month.	Not everyday	0.27	1.07	0.94	1.22
	Everyday	$<0.001^*$	1.21	1.09	1.35
Decayed, missing, or filled teeth	No	reference	-	-	-
	Yes	$<0.001^*$	0.72	0.59	0.88
Bleeding on probing (+)	No (ref)	reference	-	-	-
	Yes	0.007*	0.75	0.60	0.92
	Free caries	reference	-	-	-
	Very low	$<0.001^*$	17.51	13.61	22.53
Pocket depth > 3 mm	Low	$<0.001^*$	40.93	31.76	52.76
	Moderate	$<0.001^*$	57.80	44.63	74.86
Attachment loss > 3 mm	High	$<0.001^*$	68.80	53.05	89.24
	No	reference	-	-	-
Erosion	Yes	$<0.001^*$	1.30	1.21	1.39
	No	reference	-	-	-
Erosion	Yes	0.17	1.06	0.98	1.14
	No	reference	-	-	-
Erosion	Yes	$<0.001^*$	1.30	1.17	1.37
	No	reference	-	-	-
Erosion	Yes	0.018*	1.12	1.02	1.24
	No	reference	-	-	-

*Significant ($p < 0.05$)

Table 2 presents the results of the logistic regression analysis using both methods. People aged >60 years had 2.52 times (AOR = 2.52; 95%CI, 2.26–2.79) the number of RC compared with those aged 15–44 years. Living in rural areas (1.16 times; AOR = 1.16, 95%CI, 1.09–1.24), having a lower education (1.47 times; AOR = 1.47; 95%CI, 1.28–1.70), and never visiting a dentist (1.64 times; AOR = 1.64; 95%CI, 1.22–2.19) increased the likelihood of RC. Those who wore dentures (59%; AOR = 0.59; 95%CI, 0.48–0.73), drank carbonated water (88%; AOR = 0.88; 95%CI, 0.81–0.94), and did not brush their teeth daily (81%; AOR = 0.81; 95%CI, 0.68–0.96) were at risk of developing RC. The risk of RC was also 1.20 (AOR = 1.20; 95%CI, 1.12–1.28) or 1.21 (AOR = 1.21; 95%CI, 1.09–1.35) times higher in those who were on routine medication or smoked daily, respectively. Those who chewed betel (72%; AOR = 0.72; 95%CI, 0.59–0.88) and drank alcohol (75%; AOR = 0.75; 95%CI, 0.60–0.92) had RC. Decayed, missing, or filled teeth were shown to increase the risk of RC, where more coronal caries indicated a higher risk. Those with BOP (+) (1.30 times; AOR = 1.30; 95%CI, 1.21–1.39), attachment loss of >3 mm (1.30 times; AOR = 1.30; 95%CI, 1.17–1.37), and erosion (1.12 times; AOR = 1.12; 95%CI, 1.02–1.24) had a greater risk of developing RC.

The logistic regression showed a Hosmer–Lemeshow test value of 0.402, and the Nagelkerke R^2 value was 0.293. The results indicated a good fit for the model, as evidenced by suggesting that the model accounted for approximately 29.3% of the variability in the RC.

DISCUSSION

Root caries is a preventable disease with a good prognosis when detected early and treated promptly.⁵ It is most prevalent in older adults; this situation requires strategies or efforts for prevention and treatment by identifying the risk factors of RC. In 2021, 27.9% of Indonesia's population was aged >45 years, 39.6% were aged 20–44 years, and the average life expectancy was 73 years.⁹ From these data, it can be predicted that in the next 5–10 years, the number of older adults in Indonesia will increase sharply. This study is expected to provide an overview of the factors contributing to the occurrence of RC in Indonesian adults so that government prevention programs can be developed based on the existing conditions.

The research data were taken from the 2018 Indonesian National Health Survey,⁶ which is conducted routinely every 5 years. The data analyzed came from respondents aged ≥15 years. The results showed that the prevalence of RC was 59.4%, and on average, DMF-T was 8.56.

Our results showed that many sociodemographic factors played a role in RC occurrence, including age, residence location, and education level. The association between age and RC becomes stronger with increased age, as does the risk of caries. The highest prevalence of RC

was seen in those aged >60 years (74.2%). Compared with other countries in Asia, the rate of RC in elderly people in our sample was relatively high. For example, the RC prevalence in elderly people has been reported to be 43.9% in China¹¹ and 89.7% in Nepal.¹² In two reviews published in 2020, 5 out of 11 studies showed a positive correlation between age and RC,² as did 16 out of 25 cross-sectional studies.¹ However, several studies in the review did not report this positive correlation. The high risk of RC seen in elderly people is attributed to several factors, such as decreased ability to maintain oral hygiene, systemic diseases, increased incidence of gingival recession, and low self-efficacy.¹³ Systemic diseases that are thought to be associated with RC include cardiovascular disease and type 2 diabetes mellitus. Individuals with three or more RCs are twice as likely to experience cardiac arrhythmias, and the prevalence of xerostomia is 62% higher in patients with diabetes mellitus.¹³ The data in this study also found that taking medication regularly was associated with RC. This association may be attributed to the higher rate of systemic diseases in elderly people, which will impact the occurrence of RC and can also be triggered by the drugs they take regularly.

Participants who lived in rural areas exhibited a higher prevalence of RC, which agreed with the results of a systematic review, where two out of three cohort studies reported the same finding.² Communities in rural areas are more at risk of developing RC, possibly because access to healthcare facilities is more limited than in urban areas. The data supported this finding, as those who never visited the dentist tended to have more RCs. In terms of geography, Indonesia exhibits heterogeneity between rural and urban areas in terms of economy and distribution of healthcare facilities. As one of the most populated countries in the world,⁹ and as a lower-middle-income country with a per capita gross national income of USD 3,630, approximately half of Indonesia's population lives in rural areas with limited access to healthcare workers and a low level of welfare compared with urban areas.¹⁴ Participants with a low education level tended to have a higher prevalence of RC. By contrast, in one review, no statistical correlation was found between RC and education level. However, descriptively, people with a higher education had fewer RCs.¹ One study from Mexico reported that an increased risk of RC was not directly associated with education level.¹⁵ The authors argued that someone with higher education is typically associated with better socioeconomic and health literacy than those with low education; thus, their ability to maintain oral hygiene is also better, either independently or by visiting healthcare facilities.

Our results showed that active smokers tended to have more RCs. This finding aligns with a review reporting that smoking and RC were correlated in 5 out of 10 studies,¹ as well as results from an Australian study.¹⁰ Smoking decreases saliva's buffering ability, which can affect salivary pH and influence the concentration of salivary proteins, such as salivary secretory immunoglobulin A

and amylase. These conditions encourage an increase in the number of pathogenic bacteria in the biofilm, resulting in caries.¹⁶ This study found a strong positive correlation between coronal caries and RC. The main etiological factor in the early stages and over the course of RC in elderly people is the presence of bacteria and fermentable carbohydrates on the root surface of the teeth. Culturally, bacteria associated with RC include *Lactobacillus acidophilus* with *Streptococcus mutans*, *Actinomyces israelii* with *L. acidophilus*, and *A. israelii* with *L. acidophilus* and *S. mutans*.¹⁷ One study from Turkey found that smoking also affected the occurrence of periodontal disease,¹⁸ where the smoking group was more susceptible to advanced periodontal disease and at higher risk of bone resorption and tooth loss. There is also a link between smoking and periodontal disease and caries, as smoking triggers the formation of plaque and biofilm (i.e., the etiologies of periodontal disease and caries).

According to our regression analysis, participants who drank alcohol had significantly higher levels of RC; however, descriptively, non-alcohol drinkers tended to have higher levels of RC. In one study from China, drinking alcohol was found to be significantly correlated with increased rates of RC in a bivariate test but not in multivariate tests;¹¹ however, a study and review from Spain showed the opposite result, with no association between drinking alcohol and RC.¹⁹ Alcohol use exerts a direct biological effect on the oral cavity by altering salivary flow rate, pH value, and colonization of sub- and supragingival plaque bacterial species, as well as interfering with the immune system of the host.²⁰ However, this effect could be minimized with behaviors accompanying it to maintain good oral health, such as brushing teeth, which can reduce dental plaque. Our findings support this claim, as those who did not regularly brush their teeth were at greater risk of RC.

Although sugar is one of the main factors in the occurrence of caries, in our study, the correlation between the behaviors of drinking and eating sweets and caries development was not statistically significant, similar to other studies conducted in Bogotá, Colombia.²¹ Our study found that drinking soda was significantly associated with RC in multivariable analysis, but descriptively, respondents who did not drink soda tended to have more RC. The risk of carbonated beverages comes from their relatively high sucrose concentration, which can induce the formation of biofilms.¹⁹ Most of our study participants did not drink soda. The occurrence of RC in non-soda drinkers may be caused by other factors, such as brushing behavior, as well as the condition of the gingiva and periodontal tissue.

Respondents who chewed betel had relatively more RC. Descriptively, >50% of users of chewed betel had RC. Previous research found relationships between betel chewing and the incidence of coronal caries,²² affected periodontitis, and bone loss.⁸ Although chewing betel has an anti-cariogenic effect due to betel quid's high fluoride content and etheric oils, it also harms periodontal tissue

owing to the cholinergic impact of betel quid combined with calcium salt in the saliva, which produces calculus deposition. Heavy calculus deposition can induce the destruction of gingival tissue and the periodontal membrane.^{8,12} Chewing betel indirectly affects the occurrence of RC through the gingival and periodontal tissue, as patients with gingival or periodontal disease are at greater risk of developing RC.

Our study found that those with BOP (+) and >3 mm of attachment loss had an increased rate of RC. A separate review stated that clinical attachment loss and gingival recession are predisposing factors for RC, given the increased susceptibility of the exposed root surface.²³ Root caries generally presents as a progressive lesion found on the root of a tooth that, due to some degree of periodontal attachment loss, has been exposed to the oral environment. Studies from three different regions—England, Northern Ireland, and Wales—found that individuals with periodontitis had more coronal caries and RC.²⁴

Although erosive tooth wear and dental caries differ in their causes, both are mediated by diet. Therefore, dietary behavioral surveys can also be applied when assessing dental caries experience.²⁵ The current study found that erosion is associated with RC. Studies on adults in Norway²⁶ reported a correlation between erosion and dental caries, although they did not specifically analyze the correlation with RC. The opposite effect was observed in studies on adults in Finland.²⁵

Our results showed that people with dentures had significantly higher rates of RC. A study in Ireland found that wearing dentures was significantly associated with RC in univariate tests but not significant in multivariate tests or the final prediction model.²⁷ The effect of dentures on caries incidence is influenced by the design and type of materials used. Removable dentures have the potential to change the oral environment and increase the formation of plaque on the tooth surface covered by dentures.²⁸ Studies have shown that fixed and removable dentures increase biofilm formation and the risk of caries, gingivitis, and periodontal disease.²⁸

Based on the results of this study, it is necessary to make systematic and collective efforts between various sectors (i.e., individual, private, and government) to reduce the caries burden in all age groups.²⁹ Increasing knowledge and self-efficacy through health education in schools in collaboration with dental professionals is critical to maintaining oral health early so that coronal caries and RC can be prevented in old age.³⁰ This plan requires collaboration between the Ministry of Education and the Ministry of Health to create a synergistic curriculum. Dental professionals must be aware of the importance of early detection and prevention of RC, so its associated morbidity rate can be reduced.³¹ The government should ensure that dental professionals are equally distributed in rural areas so that people can access necessary dental services.

This study was limited in that the survey was conducted using a cross-sectional method, so the causality of the incidence and increment of RC cannot be known. In addition, regarding the variables that contribute to RC, not all variables were measured, such as economic status and intraoral conditions (e.g., oral hygiene index, gingiva recession, and level of *mutans streptococci* and *lactobacilli* in saliva). In this study, RC is based on the number of RD and RF. The number of exposed roots was not included in the observations, so it cannot be compared with roots that were exposed but did not have caries. However, the strength of this study is that the data collected came from a national survey, and the results can describe the condition of the oral health of the Indonesian population. Therefore, it is necessary to conduct an in-depth study on causality factors of RC using a longitudinal study design.

To conclude, the prevalence of RCs in Indonesia is relatively high; many occur in older adults, and most are not treated. Many different factors contribute to RC in the Indonesian adult population, including sociodemography, oral health practices, general health behaviors, and oral health status. Most of the respondents are young adults; therefore, in the next few years, the number of older adults will increase. Health behaviors that pose a risk of causing RC are still quite high among respondents, including smoking and visiting the dentist. The oral health conditions of respondents that influenced the occurrence of RC were found to be quite high, especially those related to periodontal conditions. Risk factors that can be managed through health promotion interventions to reduce the prevalence of RC are those related to behavior. The government must initiate multi-sector cooperation to design an RC prevention program, especially for older adults.

REFERENCES

1. Zhang J, Lo ECM. Epidemiology of dental root caries: a review of risk factors. *Front Oral Maxillofac Med*. 2020; 2: 5.
2. Zhang J, Sardana D, Wong MCM, Leung KCM, Lo ECM. Factors associated with dental root caries: a systematic review. *JDR Clin Transl Res*. 2020; 5(1): 13–29.
3. Hariyani N, Setyowati D, Spencer AJ, Luzzi L, Do LG. Root caries incidence and increment in the population – A systematic review, meta-analysis and meta-regression of longitudinal studies. *J Dent*. 2018; 77: 1–7.
4. López R, Smith PC, Göstemeyer G, Schwendicke F. Ageing, dental caries and periodontal diseases. *J Clin Periodontol*. 2017; 44(S18): S145–52.
5. León S, Rivera M, Payero S, Correa-Beltrán G, Hugo FN, Giacaman RA. Assessment of oral health-related quality of life as a function of non-invasive treatment with high-fluoride toothpastes for root caries lesions in community-dwelling elderly. *Int Dent J*. 2019; 69(1): 58–66.
6. Badan Penelitian dan Pengembangan Kesehatan. Laporan Nasional Riset Kesehatan Dasar 2018. Jakarta: Kementerian Kesehatan Republik Indonesia; 2018. p. 184–218.
7. Hayes M, Burke F, Allen PF. Etiology, risk factors and groups of risk. In: Carrilho MR de O, editor. *Monographs in oral science*. Karger; 2017. p. 9–14.
8. Hsiao C-N, Ting C-C, Shieh T-Y, Ko EC. Relationship between betel quid chewing and radiographic alveolar bone loss among Taiwanese aboriginals: a retrospective study. *BMC Oral Health*. 2014; 14(1): 133.
9. Badan Pusat Statistik Indonesia. Analisis profil penduduk Indonesia. Jakarta: BPS – Statistics Indonesia; 2022.
10. Hariyani N, Spencer AJ, Luzzi L, Do LG. Root caries experience among Australian adults. *Gerodontology*. 2017; 34(3): 365–76.
11. Gao YB, Hu T, Zhou XD, Shao R, Cheng R, Wang GS, Yang YM, Li X, Yuan B, Xu T, Wang X, Feng XP, Tai BJ, Hu DY, Lin HC, Wang B, Si Y, Wang CX, Zheng SG, Liu XN, Rong WS, Wang WJ, Yin W. How root caries differs between middle-aged people and the elderly: Findings from the 4th National Oral Health Survey of China. *Chinese J Dent Res*. 2018; 21(3): 221–9.
12. Joshi R, Gautam S, Joshi B. Prevalence of root caries and associated factors among elderly population visiting a dental teaching hospital in Kathmandu, Nepal. *Nepal Med Coll J*. 2023; 25(4): 341–8.
13. Gati D, Vieira AR. Elderly at greater risk for root caries: a look at the multifactorial risks with emphasis on genetics susceptibility. *Int J Dent*. 2011; 2011: 647168.
14. Wiseman V, Thabrany H, Asante A, Haemmerli M, Kosen S, Gilson L, Mills A, Hayen A, Tangcharoensathien V, Patcharanarumol W. An evaluation of health systems equity in Indonesia: study protocol. *Int J Equity Health*. 2018; 17(1): 138.
15. Islas-Granillo H, Borges-Yañez SA, Medina-Solís CE, Casanova-Rosado AJ, Minaya-Sánchez M, Villalobos Rodelo JJ, Maupomé G. Socioeconomic, sociodemographic, and clinical variables associated with root caries in a group of persons age 60 years and older in Mexico. *Geriatr Gerontol Int*. 2012; 12(2): 271–6.
16. Wu J, Li M, Huang R. The effect of smoking on caries-related microorganisms. *Tob Induc Dis*. 2019; 17(April): 32.
17. Tanner ACR, Kressler CA, Faller LL. Understanding caries from the oral microbiome perspective. *J Calif Dent Assoc*. 2016; 44(7): 437–46.
18. Beklen A, Sali N, Yavuz MB. The impact of smoking on periodontal status and dental caries. *Tob Induc Dis*. 2022; 20(August): 72.
19. Çetinkaya H, Romaniuk P. Relationship between consumption of soft and alcoholic drinks and oral health problems. *Cent Eur J Public Health*. 2020; 28(2): 94–102.
20. Huang X, Liang Y, Fan W, Liu W, Wu B, Li J. Relationship between Chinese Baijiu consumption and dental caries among 55- to 74-year-old adults in Guangdong, southern China: a cross-sectional survey. *BMC Geriatr*. 2021; 21(1): 506.
21. Usuga-Vacca M, Marin-Zuluaga DJ, Castellanos JE, Martignon S. Association between root/coronal caries and individual factors in institutionalised elderly using ICDAS severity and activity. *BMC Oral Health*. 2021; 21(1): 146.
22. Amelia AR, Batara AS, Arif AMS. The relationship between betel-chewing and dental caries: case study on Papuan college students in Makassar City. *Int J Sci Health Res*. 2020; 5(4): 263–8.
23. Bignozzi I, Crea A, Capri D, Littarru C, Lajolo C, Tatakis DN. Root caries: a periodontal perspective. *J Periodontol Res*. 2014; 49(2): 143–63.
24. AlQobaly L, Sabbah W. The association between periodontal disease and root/coronal caries. *Int J Dent Hyg*. 2020; 18(1): 99–106.
25. Alaraudanjoki V, Laitala M-L, Tjäderhane L, Pesonen P, Lussi A, Anttonen V. Association of erosive tooth wear and dental caries in Northern Finland Birth Cohort 1966 – an epidemiological cross-sectional study. *BMC Oral Health*. 2017; 17(1): 6.
26. Mulic A, Tveit AB, Skaare AB. Prevalence and severity of dental erosive wear among a group of Norwegian 18-year-olds. *Acta Odontol Scand*. 2013; 71(3–4): 475–81.
27. Hayes M, Da Mata C, Cole M, McKenna G, Burke F, Allen PF. Risk indicators associated with root caries in independently living older adults. *J Dent*. 2016; 51: 8–14.
28. Alenezi A, Alkhudhayri O, Altowaijri F, Aloufi L, Alharbi F, Alrasheed M, Almutairi H, Alanazi A, Yehya M, Al Asmari D.

- Secondary caries in fixed dental prostheses: Long-term clinical evaluation. *Clin Exp Dent Res*. 2023; 9(1): 249–57.
29. Tonetti MS, Bottenberg P, Conrads G, Eickholz P, Heasman P, Huysmans M-C, López R, Madianos P, Müller F, Needleman I, Nyvad B, Preshaw PM, Pretty I, Renvert S, Schwendicke F, Trombelli L, van der Putten G-J, Vanobbergen J, West N, Young A, Paris S. Dental caries and periodontal diseases in the ageing population: call to action to protect and enhance oral health and well-being as an essential component of healthy ageing - Consensus report of group 4 of the joint EFP/ORCA workshop on the boundaries between caries and periodontal diseases. *J Clin Periodontol*. 2017; 44 Suppl 1(S18): S135–44.
30. Shita ADP, Hamzah Z, Meilawaty Z, Indriana T, Handayani ATW, Setyowati DI. Oral health profile of the elderly people in the Pandalungan community. *Dent J*. 2020; 53(4): 217–22.
31. Qutieshat A, Salem A, Aouididi R, Delatorre Bronzato J, Al-Waeli H, Abufadalah M, Shaikh S, Yassir Y, Mhanni A, Vasantavada P, Amer H. Perspective and practice of root caries management: a multicountry study – Part I. *J Conserv Dent*. 2021; 24(2): 141–7.